

6 (a) Define *magnetic flux*.

The magnetic flux through a plane surface is the product of the component of the magnetic flux density normal to the surface and the area of the surface.

product of magnetic flux density and area [B1]

component of B normal to surface of area [B1]

[2]

(b) A solenoid has a coil C of wires wound tightly about its centre, as shown in Fig. 6.1.

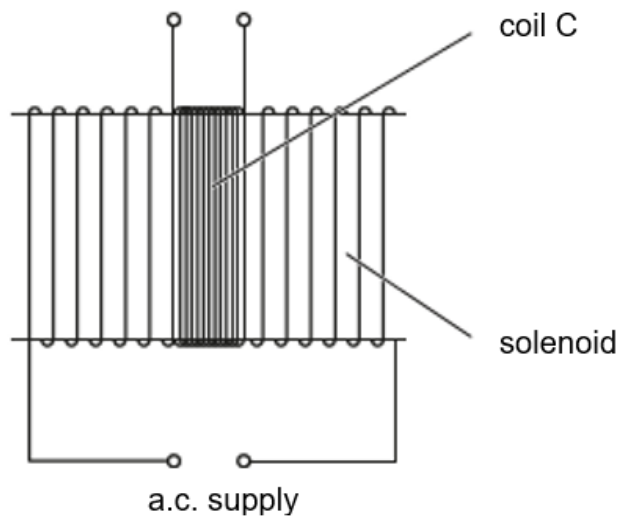


Fig. 6.1

Coil C has 86 turns.

The uniform magnetic flux Φ (in weber) in the solenoid is given by the expression

$$\Phi = 6.8 \times 10^{-6} \times I$$

where I is the current in the solenoid.

The variation with time t of the current I in the solenoid by the a.c. supply is shown in Fig. 6.2.

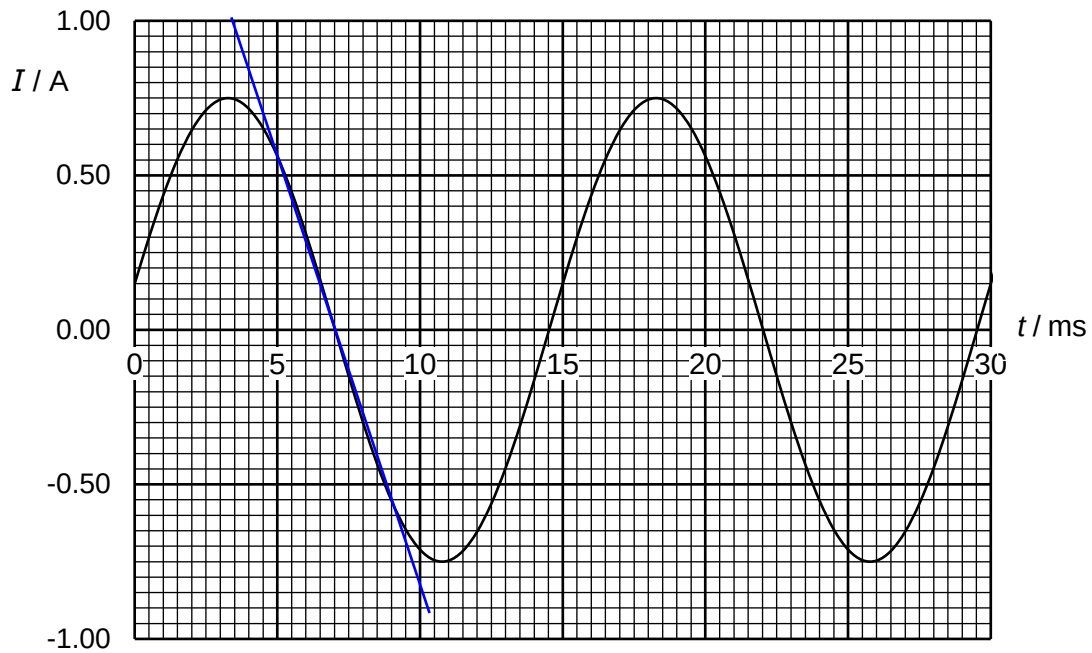


Fig. 6.2

- (i) Explain how electromotive force (e.m.f.) is induced in coil C.

As the current passes through the solenoid varies periodically, it sets up a periodical magnetic flux around the solenoid. An e.m.f. proportional to the rate of change of magnetic flux linkage across coil C will be induced in coil C.

[2]

- (ii) Use Fig. 6.2 to determine the maximum e.m.f. induced in coil C.

$$\begin{aligned}
 |\mathcal{E}_{\max}| &= N \left| \frac{d\Phi}{dt} \right|_{\max} \\
 &= 86 \times 6.8 \times 10^{-6} \times \left| \frac{1.00 - (-0.80)}{(3.5 - 10) \times 10^{-3}} \right| \\
 &= 0.16 \text{ V}
 \end{aligned}$$

e.m.f. = V [2]

(iii) The a.c. supply is changed to a non-sinusoidal periodic wave as shown in Fig. 6.3.

On Fig. 6.3, draw the corresponding e.m.f. that is induced in coil C.

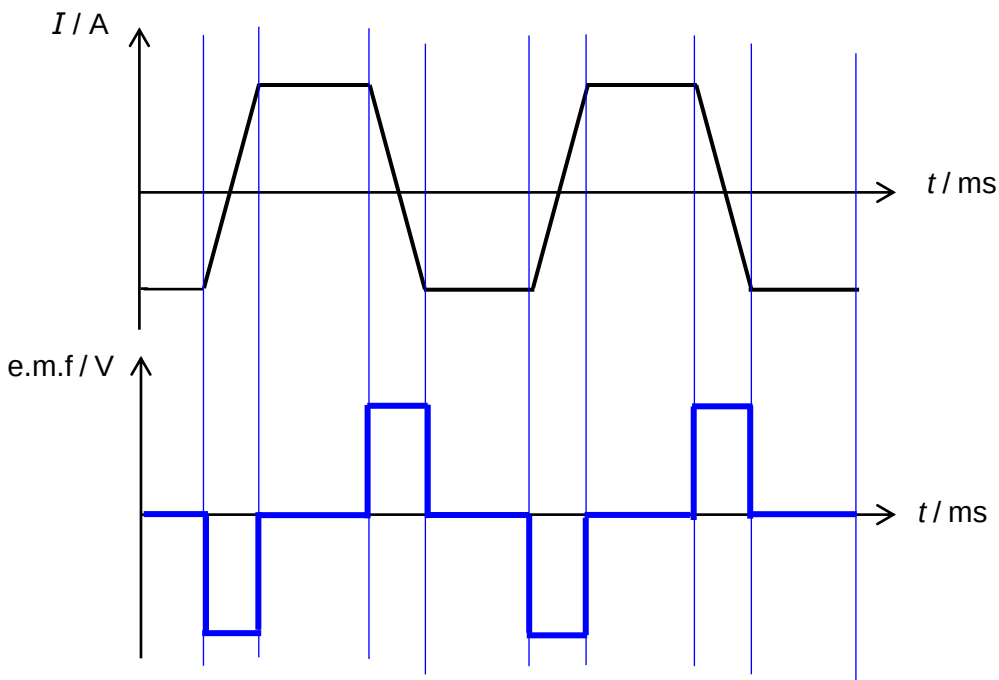


Fig. 6.3

[2]

[Total: 8]

7. (a) The radioactive isotope of Bismuth ^{210}Bi decays into Polonium (chemical symbol: Po) with